

Modeling lab, Block 2, Assignment 1

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1 Classical Thermodynamics

1.1 Question 1

The Langmuir model is based on the following three main assumptions:

1. The surface is homogeneous, meaning that all adsorption sites are equivalent and have the same adsorption energy.
2. Each adsorption site can accommodate only one molecule.
3. There are no interactions between adsorbed molecules. Therefore, the adsorption of one molecule does not affect the adsorption of molecules on neighboring sites.

1.2 Question 2

The Langmuir isotherm is given by

$$q = \frac{q_{\text{sat}} k(T) P}{1 + k(T) P}.$$

a) High-pressure limit

At high pressure, $P \rightarrow \infty$, and therefore

$$k(T)P \gg 1.$$

Thus,

$$1 + k(T)P \approx k(T)P,$$

and the Langmuir equation becomes

$$q \approx \frac{q_{\text{sat}} k(T) P}{k(T) P} = q_{\text{sat}}.$$

Therefore,

$$\lim_{P \rightarrow \infty} q = q_{\text{sat}}.$$

The gas uptake reaches a plateau because all adsorption sites are occupied and the surface is saturated.

b) Low-pressure limit

At low pressure, $P \rightarrow 0$, and therefore

$$k(T)P \ll 1.$$

Thus,

$$1 + k(T)P \approx 1,$$

and

$$q \approx q_{\text{sat}}k(T)P.$$

Hence, the gas uptake is linear with pressure:

$$q \propto P.$$

1.3 Question 3

From the low-pressure limit,

$$q \approx q_{\text{sat}}k(T)P.$$

According to Henry's law,

$$q = K_H P,$$

where K_H is the Henry coefficient.

Therefore,

$$K_H(T) = q_{\text{sat}}k(T).$$

Using the Van't Hoff equation,

$$k(T) = k_0 \exp\left(-\frac{\Delta H}{RT}\right),$$

we obtain

$$K_H(T) = q_{\text{sat}}k_0 \exp\left(-\frac{\Delta H}{RT}\right).$$

Defining $K_{H,0} = q_{\text{sat}}k_0$, this can also be written as

$$K_H(T) = K_{H,0} \exp\left(-\frac{\Delta H}{RT}\right).$$

Therefore, Henry's coefficient follows the same Van't Hoff-type temperature dependence as the equilibrium constant.

1.4 Question 4

The Langmuir model can be used to find the parameters q_{sat} and $k(T)$, using P as a variable:

$$q = \frac{q_{sat} \cdot k(T) \cdot P}{1 + k(T) \cdot P} \quad (1)$$

The reciprocal is taken:

$$\frac{1}{q} = \frac{1 + k(T) \cdot P}{q_{sat} \cdot k(T) \cdot P} \quad (2)$$

This equation can be rewritten:

$$\frac{1}{q} = \frac{1}{q_{sat} \cdot k(T)} \cdot \frac{1}{P} + \frac{1}{q_{sat}} \quad (3)$$

This equation is a linear equation $y = m \cdot x + b$ with:

$$y = \frac{1}{q}, \quad x = \frac{1}{P}, \quad m = \frac{1}{q_{sat} \cdot k(T)}, \quad b = \frac{1}{q_{sat}}.$$

Therefore, the parameter $k(T)$ can be found by identifying b and m on the linear graph $k(T) = b/m$. The parameter q_{sat} is determined with the variable b .

Schematic graph of the Equation derived from the Langmuir Equation

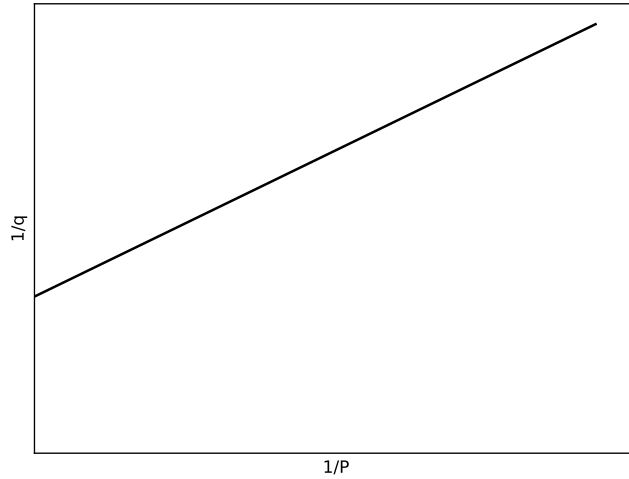


Figure 1: "Schematic graph of the Equation derived from the Langmuir Equation"

Similarly, the Van't Hoff equation can be used like this as well to determine the parameters k_0 and ΔH , with T as the variable:

$$k(T) = k_0 \cdot \exp\left(-\frac{\Delta H}{RT}\right) \quad (4)$$

By the taking the logarithmic of this equation:

$$\ln k(T) = \ln(k_0) - \frac{\Delta H}{R} \cdot \frac{1}{T} \quad (5)$$

We have obtained a linear equation of form $y = m \cdot x + b$ with:

$$y = \ln(k), \quad x = \frac{1}{T}, \quad m = -\frac{\Delta H}{R}, \quad b = \ln(k_0).$$

The parameter ΔH can be determined by finding m from the graph, with R being a known constant. The parameter k_0 can be found by taking the exponential of b .

Schematic graph of the Equation derived from the Van't Hoff Equation

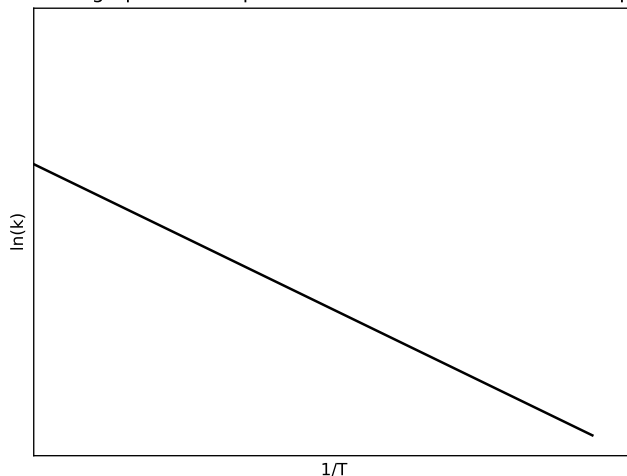


Figure 2: "Schematic graph of the Equation derived from the Van't Hoff Equation"

Note: The code used to draw the two graphs are displayed in the notebook.

1.5 Question 5

The answers to this question are on the Jupyter Notebook.

2 Statistical Thermodynamics

2.1 Question 1

In the canonical ensemble, the number of particles N , volume V , and temperature T are fixed. The system is in thermal contact with a reservoir, which fixes

its temperature.

2.2 Question 2

The canonical partition function is:

$$Q(N, V, T) = \sum_C \exp [-\beta E(C)], \quad \beta = \frac{1}{k_B T}. \quad (6)$$

2.3 Question 3

The probability of finding the system in a microstate given by E_ν is:

$$P_\nu = \frac{1}{Q(N, V, T)} \exp [-\beta E_\nu], \quad \beta = \frac{1}{k_B T}. \quad (7)$$

2.4 Question 4

2.4.1 Proof 1

First part was to prove the following equation:

$$\langle S \rangle = k_B \ln Q + k_B T \left(\frac{\partial \ln Q}{\partial T} \right)_{N, V} \quad (8)$$

Starting from the statistical definition of entropy:

$$\langle S \rangle = -k_B \sum_i p_i \ln p_i \quad (9)$$

For the canonical ensemble,

$$p_i = \frac{e^{-E_i/(k_B T)}}{Q}, \quad (10)$$

therefore

$$\ln p_i = -\frac{E_i}{k_B T} - \ln Q. \quad (11)$$

Average energy:

$$\langle E \rangle = \sum_i p_i E_i \quad (12)$$

Substituting into the entropy expression gives

$$\langle S \rangle = \frac{\langle E \rangle}{T} + k_B \ln Q. \quad (13)$$

Now,

$$Q = \sum_i e^{-E_i/(k_B T)} \quad (14)$$

and

$$\left(\frac{\partial \ln Q}{\partial T} \right)_{N,V} = \frac{1}{Q} \left(\frac{\partial Q}{\partial T} \right)_{N,V} . \quad (15)$$

Using the expression of Q in terms of p_i of expression (8) and using the definition of average energy we get:

$$\left(\frac{\partial \ln Q}{\partial T} \right)_{N,V} = \frac{\langle E \rangle}{k_B T^2}, \quad (16)$$

so that

$$\langle E \rangle = k_B T^2 \left(\frac{\partial \ln Q}{\partial T} \right)_{N,V} . \quad (17)$$

Finally, by reinjecting everything in expression (11), we get:

$$\langle S \rangle = k_B \ln Q + k_B T \left(\frac{\partial \ln Q}{\partial T} \right)_{N,V} \quad (18)$$

2.4.2 Proof 2

Second part was to prove:

$$P = k_B T \left(\frac{\partial \ln Q}{\partial V} \right)_{N,T} \quad (19)$$

Starting from the expression of the free energy for a canonical system:

$$F = -k_B T \ln Q \quad (20)$$

And using the definition of pressure:

$$P = - \left(\frac{\partial F}{\partial V} \right)_{N,T} . \quad (21)$$

We get by combining those 2:

$$P = - \left(\frac{\partial}{\partial V} [-k_B T \ln Q] \right)_{N,T} . \quad (22)$$

As T is constant we finally get:

$$P = k_B T \left(\frac{\partial \ln Q}{\partial V} \right)_{N,T} . \quad (23)$$

2.4.3 Proof 3

Third part was to prove:

$$\mu = -k_B T \left(\frac{\partial \ln Q}{\partial N} \right)_{V,T} \quad (24)$$

We also start from the expression of free energy but this time we will inject it in the definition of the chemical potential defined by:

$$\mu = \left(\frac{\partial F}{\partial N} \right)_{V,T}. \quad (25)$$

Substituting F by its expression we get:

$$\mu = \left(\frac{\partial}{\partial N} [-k_B T \ln Q] \right)_{V,T}. \quad (26)$$

Again as T is considered constant we finally get:

$$\mu = -k_B T \left(\frac{\partial \ln Q}{\partial N} \right)_{V,T} \quad (27)$$

2.4.4 Proof 4

The fourth part was to prove:

$$E = k_B T^2 \left(\frac{\partial \ln Q}{\partial T} \right)_{V,N} \quad (28)$$

Starting with the expression of the partition function:

$$Q = \sum_i e^{-E_i / (k_B T)} \quad (29)$$

and using the expression of the average energy

$$\langle E \rangle = \sum_i p_i E_i \quad (30)$$

and

$$p_i = \frac{e^{-\beta E_i}}{Q}, \quad (31)$$

We get:

$$\langle E \rangle = \frac{1}{Q} \sum_i E_i e^{-\beta E_i}. \quad (32)$$

Identifying the derivative of Q in term of β we get:

$$\langle E \rangle = \frac{1}{Q} \left(-\frac{\partial Q}{\partial \beta} \right)_{V,N} \quad (33)$$

Again we identify that this is the derivative of $\ln Q$ by β so:

$$\langle E \rangle = \left(-\frac{\partial \ln Q}{\partial \beta} \right)_{V,N} \quad (34)$$

Then by doing a change of variable β to T and thus deriving β in terms of T we get:

$$E = k_B T^2 \left(\frac{\partial \ln Q}{\partial T} \right)_{V,N} \quad (35)$$